

What the 58 say

Results, insights, limitations and the way forward, drawn from full-text extraction of 56 papers. The companion note *From 590 to 58* covers how the set was assembled.

Compiled 5 September 2026 Extracted 56 of 58 papers
Columns 9 per paper, quote-grounded Deadline 30 September 2026

Results

What the review produced, before interpretation.

590→58 Screened to shortlist	96.6% Coverage — 56 of 58 extracted	13 Papers running open-weight models	8 Papers evaluating on ARC-AGI-3
17 Grid-world or ARC-like evaluations	3 MIT-licensed reference codebases		

The clusters defined up front and the hypotheses defined separately land on a clean diagonal when crossed — A owns H3/H4, B owns H1, C owns H2/H5, E owns H6. The taxonomy is load-bearing rather than decorative. It also exposes the thin spots: **H3 has six supporting papers and H5 has seven**, and they are the only two hypotheses no ARC-AGI-3 paper has claimed.

Insights

Five things the extraction showed that were not visible from abstracts, and that change what we should build.

Working small is a method, not just a constraint

Rodionov's ablation of his own reference architecture found that *“gains from model capability and reasoning effort often exceeded differences between variants.”* At the frontier, architectural differences are drowned out by raw capability. Read as bad news, that closes the door on mechanism research. Read correctly, it says **ablations only produce signal in a constrained regime**.

Kaggle's offline sandbox forces local weights on us. That same constraint is the condition under which a mechanism comparison becomes measurable at all — which turns the project's biggest limitation into its method.

| arXiv 2607.15439 · four variants × two models × two effort levels

A better world model is not a better agent

Tycho reports that high transition accuracy **did not** improve action selection. This lands directly on H3 and H4, both of which assume that modelling the world more faithfully yields better acting. Any build resting on those hypotheses needs to measure action quality separately from model fidelity, or it will report the wrong win.

| arXiv 2607.28287 · programmatic world models, 183 levels

The field updates on contradiction, not on episode boundaries

H6 as written promotes episodic failures into semantic rules at the end of an episode. That is the minority design. Across the shortlist, far more mechanisms revise when a prediction is contradicted than when an episode ends — and PRO-LONG argues the stronger form of the objection: keep the complete log, search it with code, and do not consolidate at all.

What makes an agent change its mind



Training-loop updates are greyed because they are not an agent-visible event — they belong to how the model was fit, not to how the agent revises during play.

Among the triggers an agent can actually act on, contradiction leads by roughly three to one over episode end.

| shortlist_extracted.csv · column 3, "Update trigger"

Small models survive when decisions leave the model

SAGE wins on token- F_1 across **all seven open-weight backbones from 1B to 7B**, and the reason is structural: its novelty gate resolves ADD and NOOP in closed form, routing to an LLM only inside a narrow uncertainty band. The design rule generalises — push work out of the model into cheap deterministic components and small weights stop being fatal.

| SAGE · von Mises-Fisher novelty gate over memory embeddings

The local-weights gap is real, but narrower than we claimed

Every published *ARC-AGI-3* result runs a hosted frontier model — confirmed across all 56 rows. But 13 papers ran open-weight models in adjacent settings, so the defensible claim is not “nobody has tried small models”. It is: **the capability axis has been plotted downward in memory, exploration and episodic control — never on ARC-AGI-3.**

That is a better position. It is narrower, it is checkable, and it comes with method to borrow rather than invent.

| 13 of 56 · SAGE, Agentic Episodic Control, PREMem, Continual Harness, ReflAct, HEMA

The numbers only mean something with their protocol

Published ARC-AGI-3 scores differ by an order of magnitude, and the spread is measurement regime, not progress. RHAE is per level $(\text{human} \div \text{agent})^2$, capped at 1.15, level-weighted, per-game cap 100.

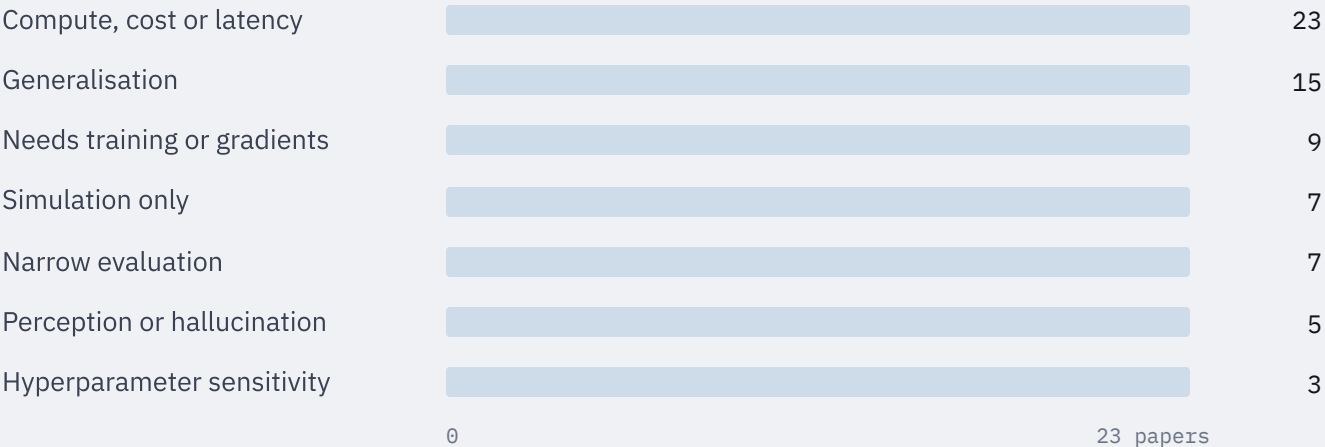
SOURCE	REPORTED	REGIME
Tycho	100.00	all 183 levels · Opus 5 / GPT-5.6 Sol
PRO-LONG	97.4%	best@2 · Fable 5 · \$1,750
Prime Agent	95.5%	best@1 · up from 30%
OPINE-World	78.4%	20/25 games · no per-game training
Rodionov ablation	34.2 – 74.8	main study · all variants and efforts
DreamTeam	38.4%	official protocol · protocol-matched · 2 runs

The two most protocol-careful papers in the shortlist are the two reporting the lowest numbers. Rodionov states the limit himself: because the newest model postdates the games and held-out performance is untested, his results “*indicate public-set saturation only.*” **DreamTeam's 38.4% is the comparable figure — not 100.**

Challenges met along the way

What the authors themselves say goes wrong

Papers naming each theme in their stated limitations, of 56



Cost is the field's own headline worry, named by 41% of the shortlist — which is exactly the axis a bounded-compute submission competes on. Only five papers name perception, but one of them, Sensi, identifies perception-layer hallucination as *its* bottleneck, and that is the failure a smaller local model would amplify.

Challenges in running the review itself

- **The citation graph was blind to its own subject.** 512 backward references and eight forward lists surfaced none of the thirteen ARC-AGI-3 papers, because no seed cites the benchmark. Found only by direct name search, twice, after the harvest looked finished.
- **Nothing was comparable until the protocol column existed.** Scores from six papers looked contradictory until each was forced to state its measurement regime.
- **The literature moved during the review.** Most of the thirteen ARC-AGI-3 papers postdate this project's start, and three arrived while the shortlist was being assembled.

Limitations of this review

Stated plainly, because a review that hides these cannot be built on. Anyone citing this work should read this section first.

- 01 **Not PRISMA.** This is snowball sampling with hand triage. There was **one rater**, no second screen, and no inter-rater reliability measure — so every inclusion and cluster decision carries a single person's judgement.
- 02 **One hop deep.** Depth-1 from 11 seeds in each direction. A two-hop review would reach further, and the classical mechanism literature is certainly under-sampled.
- 03 **The extraction is machine-generated.** Every answer is grounded in quotes from the source PDF, but the rows were spot-checked, not verified one by one. Treat any single cell as a pointer to the paper, not as evidence.
- 04 **Most evidence transfers by analogy.** Only 17 of 56 papers evaluate on grid-world or ARC-like environments, and only 8 touch ARC-AGI-3. Conclusions about what will work on this

benchmark are inference, not observation.

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- 05 **The shortlist is quota-shaped.** Cluster sizes reflect our quota decisions as much as the literature's real distribution. That cluster E holds 21 papers and D holds 4 is partly a fact about memory research and partly a choice we made.
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- 06 **A snapshot, with two holes.** Current to 5 September 2026. Two papers were unreachable — one paywalled, one with no resolvable identifier — so coverage is 56 of 58. Both are acquisition failures, not screening decisions.
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Way forward

In priority order. The ordering is the recommendation.

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- 01 **Clear the external dependency**
- Kaggle account and phone verification gate GPU access and are the only outstanding item that someone else controls. Everything below assumes they are done.
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- 02 **Run the capability sweep nobody has run**
- Hold the MIT-licensed reference architecture fixed and sweep model capability downward to local weights, reporting protocol-matched RHAЕ at each point. This is the contribution the review actually points to: untried on this benchmark, and the regime where mechanism differences stop being masked by raw capability.
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- 03 **Read three papers, not fifty-eight**
- `arc-3-agents-baseline1` — code before paper — then Tycho and OPINE-World. Those three carry the architecture, the ceiling result and the strongest exploration mechanism. The other 55 inform; these decide.
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- 04 **Test H3 or H5, with the caveat attached**

They are the only hypotheses unclaimed on ARC-AGI-3, but they are also the thinnest in our own shortlist — six and seven papers. Thin coverage means open ground *or* a dead end, and Tycho's finding that model fidelity did not improve action selection is a live threat to H3. Decide by reading, not by novelty.

05

Design for the small model from the first line

Follow SAGE: put every decision that can be closed-form into deterministic code, and spend LLM calls only where genuine ambiguity remains. Retro-fitting this after building against a frontier API means writing the agent twice.

The one-sentence version. Efficiency on ARC-AGI-3 is no longer a novel angle, but the published numbers are frontier-model, best-of- k numbers on a saturating public set — and the axis nobody has swept is the one Kaggle forces us onto anyway.

That is a narrow claim, it is checkable, and unlike the version we started with, it survives contact with the literature.

Drawn from `research_prep/data/shortlist_extracted.csv` — 58 rows, 57 columns, 56 papers extracted across nine quote-grounded fields. Method and provenance in the companion note *From 590 to 58*. Figures current to 5 September 2026.